

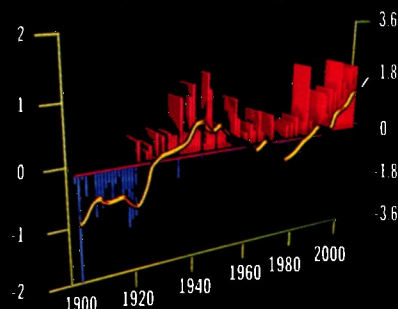
ARCTIC WARNING

The Arctic Circle including Greenland, which plays a critical role in regulating the earth's climate, is now under threat. Here's why.

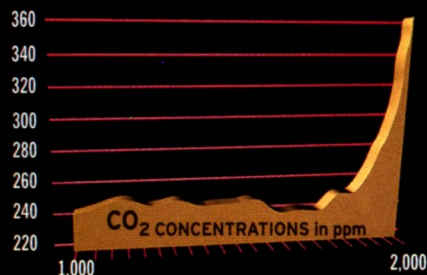
2 Arctic warms faster than lower latitudes because as the ice melts, the darker land and ocean surfaces absorb more solar energy that results in a cycle of warming



3 Arctic temperatures have been rising by over 2-3 degrees C

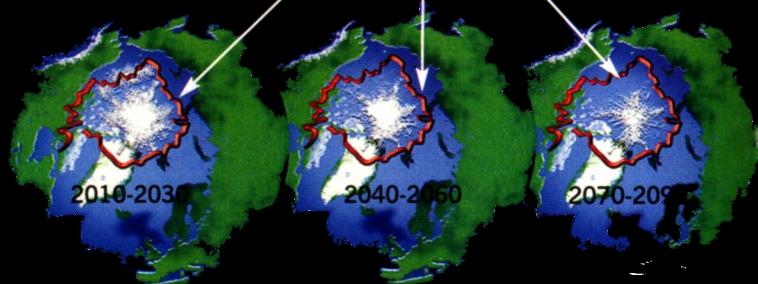


1 CO₂ levels are 28 % higher than 800 years ago and rising 200 times faster



4 Arctic sea ice is shrinking

Observed Ice Extent
Sept 2002



Projected Sea Ice Extent in the Arctic

observed earlier that evening (according to my watch) as our waterproof boots sank deep into the soft snow. We are decked with thermal wear that makes us feel like, well, Polarnauts, with our heads well-covered to keep our ears from freezing. We had dark sunglasses to avoid sore eyes from the disorienting dazzle of sunlight reflecting off the snow.

We watch three scientists laboriously mount an ice core driller in the middle of nowhere. It is being supervised by Sigfus Johnsen who has been working on driller cores for the past 32 years, including a stint in Antarctica, and has lost count of the number of

bores he has made. Yet under his bright blue hood, his eyes light up as the team pulls out a cylindrical core of ice about a metre long. Several samples are then carted on a skidoo to an underground ice laboratory, a kilometre away for analysis.

Later at the geodesic-shaped main hall, where the temperatures are toasty, some of the ice core samples are cut up by the team to have whisky on the rocks. Johnsen says bemusedly, "That ice you are sipping is 123 years old." The year these formed coincided with the birth of the Indian National Congress in 1885. The cubes have tiny

bubbles of air that let off a popping sound. When analysed, they contain authentic information of the climatic conditions that existed in the year of their formation including the temperature and the levels of carbon dioxide in the atmosphere. Similar to what the rings in the bark of a tree reveal about its age and the conditions it experienced.

For the research team drawn from 14 nations (India is a surprise absentee), the ice core they have just extracted is a relative toddler compared to their quest. They want core samples from a depth of 2,500 metre, which when dissected would reveal what climatic